

3.1.1 Sequences Worksheet

Determine whether each of the following sequences converges or diverges.

1. $\left\{ \frac{2n}{3n-1} \right\}_{n=1}^{\infty}$

2. $\{6-3n\}_{n=1}^{\infty}$

3. $\left\{ \frac{n}{e^n} \right\}_{n=1}^{\infty}$

4. $\left\{ \sqrt[n]{n} \right\}_{n=1}^{\infty}$

$$5. \quad \left\{ (-1)^n \frac{1}{n} \right\}_{n=1}^{\infty}$$

$$6. \quad \left\{ (-1)^n \frac{4n}{1-7n} \right\}_{n=1}^{\infty}$$

$$7. \quad \left\{ (-1)^n \frac{1}{4^n} \right\}_{n=1}^{\infty}$$